Software Test Plan (STP)

Project: Event Management System

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1. Introduction

1.1 Purpose

This Software Test Plan (STP) defines the testing objectives, scope, strategy, environment, resources, schedule, deliverables, risks, and traceability approach for the Event Management System (EMS). It is derived from the approved/planned requirements in the EMS Software Requirements Specification (SRS) and the corresponding Software Architecture and Design Specification (SAD).

1.2 Scope

Testing covers the web-based Event Management System functions for Admin, Organizer, Attendee, and Event Staff. The scope includes authentication and RBAC, event lifecycle management, venue/capacity handling, event discovery, registration, payment-status recording, ticket/QR generation and validation, attendance, notifications, feedback, reporting, administration, audit logging, performance, usability, reliability, and security controls.

Internal implementation of external payment gateways, email/SMS providers, and third-party authentication providers is outside the product scope; where applicable, EMS adapter behavior and failure handling will be tested.

1.3 References

- Event Management System Software Requirements Specification (SRS), Version 1.0, dated 05-09-2026.
- Event Management System Software Architecture and Design Specification (SAD), Version 1.0, dated 14-09-2026.
- Supplied Software Test Plan template for Software Engineering.
- UML/architecture and API design contained in the SAD.

1.4 Definitions

Term	Definition
EMS	Event Management System.
SRS	Software Requirements Specification.
SAD	Software Architecture and Design Specification.
STP	Software Test Plan.
RTM	Requirements Traceability Matrix.
RBAC	Role-Based Access Control.
QR	Quick Response code used for digital event passes.
UAT	User Acceptance Testing.
TC	Test Case.
API	Application Programming Interface.
TLS	Transport Layer Security.

2. Test Items

- Responsive Web UI and role-specific dashboards.
- API / Controller Layer.

- Authentication and RBAC module.
- Event Management module.
- Registration and Capacity module.
- Payment Status Adapter.
- Ticket / QR module.
- Attendance and check-in module.
- Notification module.
- Feedback and Reporting modules.
- Administration and Audit module.
- Repository / Relational Database layer.
- Integration boundaries for payment, notification, and optional authentication providers.

3. Features to be Tested

The following features are mapped to the requirement IDs defined in the SRS and the design/test references carried into the SAD.

Req ID	Feature / Requirement	Priority	Primary Test Ref
EMS-F-001	User registration	High	TC-REG-01
EMS-F-002	Authentication and role-based access	High	TC-AUTH-01
EMS-F-003	Create event	High	TC-EVT-01
EMS-F-004	Edit/publish/postpone/cancel event	High	TC-EVT-02
EMS-F-005	Search and filter events	Medium	TC-SEARCH-01
EMS-F-006	Display complete event details	High	TC-EVT-03
EMS-F-007	Event registration when seats are available	High	TC-REG-02
EMS-F-008	Capacity enforcement / event full state	High	TC-REG-03
EMS-F-009	Payment status recording	High	TC-PAY-01
EMS-F-010	Generate ticket / QR pass	High	TC-TKT-01
EMS-F-011	Ticket / QR validation and check-in	High	TC-CHK-01
EMS-F-012	Registration/reminder/update/cancellation notifications	Medium	TC-NOTIF-01
EMS-F-013	Attendee list and registration status	High	TC-LIST-01
EMS-F-014	Post-event feedback and ratings	Medium	TC-FB-01
EMS-F-015	Event/registration/attendance/payment reports	High	TC-REP-01
EMS-F-016	Admin management	High	TC-ADMIN-01
EMS-F-017	Audit record	High	TC-AUD-01
EMS-NF-001	95th percentile normal response time ≤ 3 seconds	High	TC-PERF-01
EMS-NF-002	Monthly availability ≥ 99.5% excluding scheduled maintenance	High	TC-REL-01
EMS-NF-003	Protection of personal data / no card credential storage	High	TC-SEC-01
EMS-NF-004	At least 500 concurrent users	Medium	TC-SCALE-01

EMS-NF-005	Responsive browser usability and clear validation/errors	Medium	TC-UX-01
EMS-NF-006	Timestamped operational/security logs	High	TC-OPS-01

4. Features Not to be Tested

- Internal implementation of external payment gateway processing; only the EMS payment-status adapter, state handling, and failure behavior are in scope.
- Internal implementation of external email/SMS/notification providers; EMS notification generation and failure handling are in scope.
- Internal implementation of third-party authentication providers; EMS authentication boundary and authorization behavior are in scope.
- Vendor-side infrastructure or firmware of external services.
- Production-scale operational monitoring beyond the deployment/test environment unless explicitly provided for the course.

5. Test Approach / Strategy

5.1 Test Levels

- Unit Testing – verify individual services/modules such as authentication, registration, capacity checks, ticket generation, and reporting logic.
- Integration Testing – verify interactions among the API layer, modules, repository/database, and external-service adapters.
- System Testing – verify complete end-to-end workflows in the EMS web application.
- Acceptance Testing (UAT) – verify that representative user workflows satisfy the SRS acceptance criteria.

5.2 Test Types

- Functional testing for all high- and medium-priority functional requirements.
- Regression testing after defect fixes or changes to shared modules.
- Performance and load testing for response time and the 500-concurrent-user target.
- Reliability testing for failure/retry handling and transaction consistency.
- Usability testing on supported desktop and mobile browser sizes.
- Security testing covering TLS, authentication, RBAC, input validation, session expiry, and secure logging.
- Negative testing for invalid credentials, unauthorized roles, full events, invalid/reused QR passes, failed payments, malformed input, and unavailable integrations.

5.3 Entry Criteria

- A stable test build is available.
- Required test environment and database are available.
- Required test data and representative user accounts are prepared.

- Relevant SRS/SAD requirements and acceptance criteria are baselined.
- Test cases for the planned test cycle are reviewed.

5.4 Exit Criteria

- 100% of planned test cases for the cycle are executed.
- No Critical/Blocker defects remain open.
- All mandatory high-priority acceptance criteria are satisfied.
- Security requirements in scope pass validation or have formally approved deviations.
- Performance and scalability targets are measured and documented.
- A test summary report is prepared with pass/fail status, defects, coverage, and outstanding risks.

5.5 Security Validation

Security Area	Validation	Test Ref
TLS / HTTPS	Verify client-server communication uses HTTPS with TLS 1.2+ where deployed.	TC-SEC-01
Password handling	Verify passwords are stored only as strong salted hashes and are not stored in plaintext.	TC-SEC-02
RBAC	Verify Admin, Organizer, Attendee and Event Staff cannot access unauthorized operations.	TC-SEC-03
Input validation	Use malformed and malicious inputs to verify safe rejection/handling and reduced injection risk.	TC-SEC-04
Session security	Verify inactive authenticated sessions expire and logout invalidates the session.	TC-SEC-05
Audit logging	Verify important security/admin events are timestamped and logs do not contain passwords or unnecessary secrets.	TC-SEC-06

6. Test Environment

Environment Area	Configuration / Requirement
Client	Modern desktop and mobile browsers; responsive UI test at representative screen sizes.
Application	EMS server-side web/API application test build.
Database	Relational DBMS configured for transactional persistence and capacity-sensitive operations.
Network	HTTPS/TLS test configuration; controlled network failure scenarios where feasible.
API Testing	REST-style endpoints specified in the SAD, including authentication, events, registration, ticket, attendance and reporting APIs.
External Integrations	Payment, notification and optional authentication test stubs/sandboxes where available.
Performance	Load-test environment capable of generating up to at least 500 concurrent users.
Test Data	Dummy users for all four roles, events with different capacities/statuses, registrations, payment states, tickets/QR passes, attendance records, feedback and report data.
Defect Tracking	A team-approved issue tracker or repository issue mechanism, if required by the course/project process.

The SAD intentionally keeps the technology stack vendor-neutral because the SRS specifies technology categories rather than exact products. Therefore, this STP does not invent a specific framework, database vendor, cloud provider, or testing tool.

7. Test Schedule

Milestone	Start	End	Activity
Test-plan preparation	15-Sep-2026	15-Sep-2026	Finalize STP and traceability.
Test-case design/review	16-Sep-2026	17-Sep-2026	Prepare and review functional, security and non-functional cases.
Environment setup	18-Sep-2026	18-Sep-2026	Prepare build, database, test data and integration stubs.
Functional test execution	19-Sep-2026	21-Sep-2026	Execute functional and negative tests.
Integration/system testing	22-Sep-2026	23-Sep-2026	Execute end-to-end and integration workflows.
Security validation	24-Sep-2026	24-Sep-2026	Validate TLS, RBAC, input validation, sessions and audit logging.
Performance/scalability testing	25-Sep-2026	25-Sep-2026	Measure response time and 500-user target.
Regression testing	26-Sep-2026	27-Sep-2026	Retest fixes and affected workflows.
UAT / final verification	28-Sep-2026	29-Sep-2026	Representative user acceptance and final evidence.
Test summary	30-Sep-2026	30-Sep-2026	Finalize results, metrics, defects and sign-off.

These dates are a proposed project testing schedule for the STP and should be changed if your course has an official submission/execution schedule.

8. Test Deliverables

- Software Test Plan (this document).
- Test case specification covering functional, security, performance, usability and negative scenarios.
- Test data / test accounts.
- Test execution results or logs.
- Defect reports, where defects are found.
- Requirements Traceability Matrix / test coverage mapping.
- Performance and scalability test evidence.
- Security validation evidence.
- Final Test Summary Report.

The reference STP template lists these as test deliverables. Whether every execution artifact must be submitted separately is not specified in the supplied documents; follow the course faculty's submission instructions if provided.

9. Roles and Responsibilities

Role	Name	Responsibility
Test/QA Coordinator	Vineet Toshniwal	Coordinate STP, test-case review, execution tracking and final test summary.
Test Engineer	Vankadara Sri Kruthi	Design/execute assigned functional and usability cases; record evidence and defects.
Test Engineer	Vikhyath Agrawal	Execute integration, API, performance and regression tests; maintain test results.
Developer / Test Support	Divyanshu Jha	Support environment/build setup, defect reproduction and fixes; assist with security/integration validation.
Course/Project Guide	<Name>	Review test plan and provide acceptance/sign-off as required.

10. Risks and Mitigation

Risk	Severity	Mitigation
Unstable build delays testing	Medium	Request early stable/smoke builds; prioritize blocked high-priority cases.
Database/environment unavailable	Medium	Maintain a reproducible test setup and backup test data.
External payment/notification provider unavailable	Medium	Use test stubs/sandboxes; validate adapter failure handling independently.
Concurrent registration causes capacity inconsistency	High	Test atomic capacity enforcement under concurrent requests.
Unauthorized access through role manipulation	High	Perform server-side RBAC tests for every protected role boundary.
Sensitive data appears in logs	High	Inspect audit/application logs and reject builds that expose passwords, tokens or unnecessary secrets.
Performance target cannot be measured reliably	Medium	Use a controlled load-test environment and record configuration, workload and percentile results.
Insufficient test data	Low	Prepare representative users, events, registrations, payments, tickets and attendance records before execution.

11. Assumptions & Dependencies

- A testable EMS build implementing the SRS/SAD scope will be made available.
- Test accounts for Admin, Organizer, Attendee and Event Staff can be created.
- A relational database test instance is available for transactional and data-integrity testing.
- Payment and notification integrations can be stubbed/sandboxed when real external providers are unavailable.
- Test data can be reset between test cycles.
- Browser-based testing can be performed on representative desktop and mobile screen sizes.

- The final implementation may choose specific technologies later; test execution will be adapted to the selected implementation.
- The exact course submission mechanism is controlled by the course/project instructions and is not defined in the supplied SRS, SAD or sample STP.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for a prolonged period and blocks planned execution.
- The build is sufficiently unstable that a significant portion of planned test cases cannot be executed.
- A Critical/Blocker defect prevents meaningful testing of core workflows.
- Required test data or integration stubs are unavailable.

Resume testing when:

- Blocking defects are resolved or an approved workaround is available.
- The test environment and build are stable.
- Required test data and dependencies are restored.
- The affected test cases are re-planned or re-baselined where necessary.

13. Test Case Management & Traceability

The STP uses the requirement IDs and test references already established in the SRS/SAD. The SAD maps each requirement to a design specification and a test reference, providing the baseline for the test-case matrix.

Test Ref	Requirement	Area	Scenario	Expected Result
TC-REG-01	EMS-F-001	Registration	Valid registration data creates account	Account created + confirmation
TC-AUTH-01	EMS-F-002	Authentication/RBAC	Valid/invalid credentials and role dashboard	Correct access / rejection
TC-EVT-01	EMS-F-003	Event	Create event with mandatory fields	Event saved
TC-EVT-02	EMS-F-004	Event	Edit/publish/postpone/cancel authorized event	Status/change reflected
TC-SEARCH-01	EMS-F-005	Search	Search/filter by category/date/venue/availability	Matching events shown
TC-EVT-03	EMS-F-006	Event Details	Verify displayed event details are current	Accurate details shown
TC-REG-02	EMS-F-007	Registration	Register when capacity exists	Unique registration + capacity reduced

TC-REG-03	EMS-F-008	Capacity	Attempt registration at zero availability	Registration rejected + full state
TC-PAY-01	EMS-F-009	Payment	Verify successful and failed payment states	Paid confirmed / failure not confirmed
TC-TKT-01	EMS-F-010	Ticket/QR	Generate pass after confirmed registration	Unique ticket/QR created
TC-CHK-01	EMS-F-011	Attendance	Validate valid, invalid and reused pass	Valid check-in / invalid rejected
TC-NOTIF-01	EMS-F-012	Notification	Trigger registration/update/reminder/cancellation notification	Notification generated
TC-LIST-01	EMS-F-013	Lists	Authorized user views participant list	Correct filtered list
TC-FB-01	EMS-F-014	Feedback	Eligible attendee submits feedback once	Feedback saved
TC-REP-01	EMS-F-015	Reporting	Generate report by event/date/type	Report returned
TC-ADMIN-01	EMS-F-016	Admin	Admin manages users/categories/venues/config	Authorized changes saved
TC-AUD-01	EMS-F-017	Audit	Perform tracked admin/event action	Actor/time/action recorded
TC-PERF-01	EMS-NF-001	Performance	Measure normal request latency	95th percentile $\leq$ 3s
TC-REL-01	EMS-NF-002	Reliability	Review availability/error handling	Target $\geq$ 99.5% excluding scheduled maintenance
TC-SEC-01	EMS-NF-003 / EMS-SR-001	Security	Verify transport/data protection	TLS + sensitive data protection pass
TC-SCALE-01	EMS-NF-004	Scalability	Load test with 500 concurrent users	No critical failure
TC-UX-01	EMS-NF-005	Usability	Test representative browsers/screen sizes	Usability criteria pass
TC-OPS-01	EMS-NF-006	Auditability	Inspect required timestamped logs	Required events recorded
TC-SEC-02	EMS-SR-002	Security	Inspect password storage	Strong salted hashes; no plaintext
TC-SEC-03	EMS-SR-003	Security	Attempt cross-role privileged access	Unauthorized access denied
TC-SEC-04	EMS-SR-004	Security	Submit malformed/malicious inputs	Rejected/safely handled
TC-SEC-	EMS-SR-	Security	Test inactivity timeout and logout	Protected

05	005			resource inaccessible
TC-SEC-06	EMS-SR-006	Security	Inspect audit logs for secrets and required fields	No credentials; required events present

14. Test Metrics & Reporting

- Test case execution percentage = executed / planned × 100.
- Pass percentage and fail percentage.
- Requirement coverage = requirements with mapped/executed tests / total in-scope requirements × 100.
- Defect count by severity and status.
- Defect density, where a meaningful implementation size measure is available.
- Defect aging for unresolved defects.
- 95th percentile response time for normal requests.
- Maximum supported concurrent users achieved without critical failure.
- Security requirement pass/fail status.
- UAT acceptance status.

14.1 Reports

- Test execution status during the test cycle.
- Defect/issue report.
- Performance and scalability results.
- Security validation results.
- Final Test Summary Report with requirement coverage and outstanding risks.

15. Approvals

Role	Name	Signature / Date
QA / Test Coordinator		_____
Project Guide	<Name>	_____
Course Coordinator	<Name>	_____